Siddhant Agarwal

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Summary

Full-stack software engineer with 2+ years of production C#/.NET experience building distributed, fault-tolerant systems at scale — currently targeting roles in defense, aerospace, and high-reliability software engineering.

Education

Rensselaer Polytechnic Institute — Troy, NY
Bachelor of Science in Computer Science, Minor in Cognitive Science of AI — *Aug 2021 – May 2025*

- GPA: 3.53 | Dean's Honor List | U.S. Citizen — Eligible for Security Clearance
- **Relevant Coursework:** Cloud Computing, Operating Systems, Distributed Systems, Database Systems, Data Structures & Algorithms, Software Design, Machine Learning, Programming Languages

Technical Skills

Languages: C/C++, C#, Python, Java, JavaScript, SQL (MySQL, PostgreSQL)
Frameworks & Tools: .NET, Blazor, React, FastAPI, AWS, Docker, Redis, Git, Linux, TensorFlow, PyTorch
Concepts: Distributed Systems, Real-Time Systems, Concurrent & Parallel Programming, Fault Tolerance, Backend Engineering, System Design, Performance Optimization, Software Reliability, REST APIs, Testing & Debugging

Experience

Software Developer — Jul 2025 – Present
Tenex Software Solutions — *Tampa, FL*

- Owned backend development on a mission-critical election platform — designed voter data schemas, built an idempotent SQL migration pipeline, and implemented a role-based permission system spanning three administration modules
- Delivered Election Link end-to-end: designed the DB schema, shipped 10 voter-facing features (vote-by-mail, provisional ballot, sample ballot, polling location finder), and architected a DB-driven component system enabling zero-code deploys across 2,300+ county sites
- Hardened the platform component library: fixed a VoterID session-restore race condition across 9 components, shipped full keyboard accessibility (arrow-key navigation + MutationObserver) across 9 pages, and built a configurable dropdown filter adopted across all grid views
- Resolved 4 performance bottlenecks across the voter portal: parallelized page loads with Task.WhenAll, eliminated N+1 DB queries with batch IN calls, restructured voter search SQL for MySQL index pushdown, and built a skeleton loading system eliminating blank-screen flash on first render

Software Developer Intern — Jan 2024 – Jul 2024
Tenex Software Solutions — *Tampa, FL*

- Shipped 50+ features and defect fixes across a production .NET election platform, resolving data-layer bugs and UI regressions that directly impacted election workflow correctness
- Optimized 20+ PostgreSQL queries and redesigned 10 UI components, improving data retrieval performance by 40% under production workloads
- Built internal automation tooling that replaced 3+ recurring manual ballot tracking workflows, standardizing data pipelines and reducing election-day reporting overhead

Undergraduate Teaching Assistant — Sep 2023 – Dec 2023
Rensselaer Polytechnic Institute — *Troy, NY*

- Led weekly office hours for 20+ students, reinforcing algorithms and data structures through structured problem-solving and guided debugging
- Graded exams for 100+ students covering trees, graphs, dynamic programming, and greedy algorithms, providing detailed technical feedback on correctness and efficiency

Software Engineer Intern — Jun 2022 – Aug 2022
ThoughtFocus Inc. — *Edison, NJ*

- Integrated LLM and NLP models (BERT, GPT-based) into enterprise workflows, reducing manual document processing time by 20% through automation
- Developed automated test suites using PyTest and Selenium to validate end-to-end enterprise workflows, reducing production defect leakage by 80%

Projects

Real-Time 2D Game Engine (SFML) | *C++, Systems Design* | github — Sep 2025 – Present

- Engineered a real-time 2D game engine in C++ using SFML targeting 60+ FPS with 200+ active entities, implementing an event-driven architecture and multi-layer rendering pipeline with sub-16ms frame budgets
- Implemented performance-critical modules including physics, collision detection, and asset loading with spatial partitioning, reducing collision checks from $O(n^2)$ to $O(n \log n)$

Explainability of AI MIT App Inventor Project | *Java, Python* | github — Sep 2024 – Mar 2025

- Converted neural network outputs into decision-tree approximations, reducing inference complexity from $O(n)$ to $O(\log n)$ for resource-constrained deployments
- Resolved Android SDK and build configuration issues across multiple environments to ensure stable deployment

Text Generation Training Model | *Python, TensorFlow* | github — Aug 2023 – Aug 2024

- Developed an LSTM-based sequence model trained on 10,000+ movie plots, achieving 78% next-token prediction accuracy on held-out validation data
- Built preprocessing pipelines for tokenization and sequence modeling, reducing training data preparation time by 35%

Leadership

Technology Director / Events Co-Director — Apr 2022 – Apr 2025
Society of Asian Scientists and Engineers — *Troy, NY*

- Led 7-member team to architect and deploy AWS EC2-hosted organizational websites, configuring infrastructure and increasing visitor traffic by 70%
- Organized 4+ monthly technical workshops and 2 large-scale networking events with 100+ attendees, coordinating logistics, speakers, and scheduling